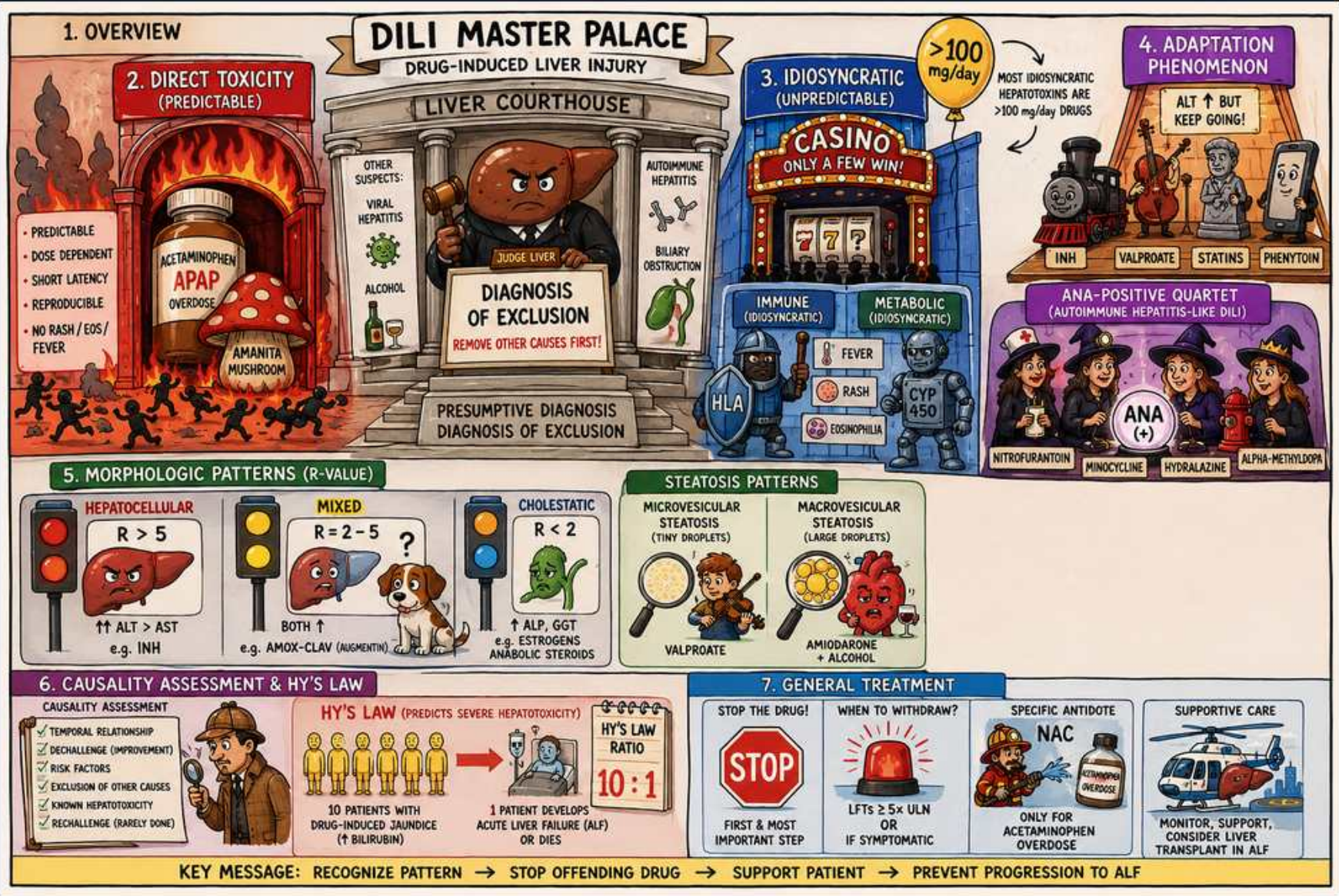


DILI — Master Palace: Patterns, Hy's Law & Treatment



1. DILI Master Palace banner

WHAT YOU SEE

DILI MASTER PALACE — Drug-Induced Liver Injury sign

WHAT IT ENCODES

Master overview of drug-induced liver injury, the leading cause of acute liver failure in the US and the #1 reason drugs are pulled from market. Organize every DILI question along four axes: mechanism (intrinsic/direct vs idiosyncratic), morphologic pattern (R-value: hepatocellular vs cholestatic vs mixed), severity (Hy's Law), and management (stop the drug ± NAC). Remember DILI is always a diagnosis of exclusion.

BOARD TRAP

BOARD TRAP: a question giving a clear drug exposure plus elevated LFTs wants you to anchor on the drug, but you must first exclude viral hepatitis (A/B/C/E), autoimmune hepatitis, and biliary obstruction (RUQ ultrasound) before calling it DILI.

2. Direct toxicity (predictable) hall

WHAT YOU SEE

Room 2: DIRECT TOXICITY (PREDICTABLE), APAP overdose, Amanita mushroom

WHAT IT ENCODES

Intrinsic/direct DILI is dose-dependent, predictable, reproducible, and has a short latency (hours to days) because the toxin or its reactive metabolite directly injures hepatocytes. Prototypes: acetaminophen (>10 g or >150 mg/kg; toxic metabolite NAPQI), Amanita phalloides, carbon tetrachloride. Because it is dose-related it is reproducible in animal models, unlike idiosyncratic injury.

BOARD TRAP

WRONG to call intrinsic DILI 'unpredictable' or 'host-dependent' — predictability and dose-dependence are the defining features that distinguish it from idiosyncratic injury.

3. Diagnosis of exclusion judge

WHAT YOU SEE

Judge in Liver Courthouse: DIAGNOSIS OF EXCLUSION, remove other causes first

WHAT IT ENCODES

DILI is a presumptive clinical diagnosis of exclusion — there is NO confirmatory test. Establish a temporal relationship to drug start, then rule out viral hepatitis (HAV IgM, HBsAg/anti-HBc IgM, HCV RNA, HEV), autoimmune hepatitis (ANA/ASMA), and biliary obstruction (RUQ ultrasound/MRCP). RUCAM/Roussel-Uclaf score formalizes causality.

BOARD TRAP

WRONG answer: ordering a liver biopsy 'to confirm DILI' — biopsy is supportive/nonspecific and the diagnosis is clinical; missing an acute viral or obstructive cause is the real danger.

4. Idiosyncratic casino

WHAT YOU SEE

Room 3: IDIOSYNCRATIC (UNPREDICTABLE) CASINO, only a few win

WHAT IT ENCODES

Idiosyncratic DILI is rare (roughly 1 in 1,000 to 100,000 exposed), host-dependent, and not clearly dose-dependent, with variable latency of 5–90 days. Two subtypes: immune/hypersensitivity (HLA-linked, fever/rash/eosinophilia, e.g., DRESS) and metabolic (reactive metabolite via CYP450). Its rarity is why it escapes pre-marketing clinical trials.

BOARD TRAP

WRONG to expect idiosyncratic DILI to be dose-proportional or to appear within hours — its variable weeks-to-months latency and lack of dose dependence are the discriminators from intrinsic injury.

5. >100 mg/day risk note

WHAT YOU SEE

Sign: most idiosyncratic from >100 mg/day drugs

WHAT IT ENCODES

The 'Rule of 2': drugs given at >50–100 mg/day AND with high lipophilicity (significant hepatic metabolism) carry the highest idiosyncratic DILI risk, because more reactive metabolite is generated. Most idiosyncratic offenders are dosed above 100 mg/day.

BOARD TRAP

WRONG to conclude a low daily-dose drug 'cannot' cause idiosyncratic DILI — dose only shifts probability; idiosyncratic injury is still possible at low doses, unlike the strict dose threshold of intrinsic toxins.

6. Immune (HLA) vs metabolic (CYP450)

WHAT YOU SEE

IMMUNE box (HLA, eosinophils) vs METABOLIC box (CYP450)

WHAT IT ENCODES

Idiosyncratic mechanisms split into immune-mediated (HLA-restricted T-cell response; fever, rash, eosinophilia, short re-exposure latency, e.g., DRESS, abacavir/HLA-B*57:01, flucloxacillin) versus metabolic (CYP450 generates a reactive metabolite that depletes glutathione / forms adducts; no allergic features). Immune type recurs rapidly on rechallenge; metabolic type does not.

BOARD TRAP

WRONG to demand fever/rash/eosinophilia before diagnosing idiosyncratic DILI — the metabolic subtype (and INH hepatitis) has NO allergic features, so their absence does not exclude DILI.

7. Adaptation phenomenon train

WHAT YOU SEE

Room 4: ADAPTATION PHENOMENON, ALT up but KEEP GOING (INH, valproate, statins, phenytoin)

WHAT IT ENCODES

Adaptation = a transient, asymptomatic ALT rise (usually <3x ULN) that resolves despite CONTINUING the drug, due to hepatocellular tolerance. Classic with isoniazid, statins, valproate, and phenytoin. Not every enzyme elevation mandates stopping — monitor when mild and asymptomatic.

BOARD TRAP

WRONG to stop an adaptation-prone drug (e.g., a statin or INH) for an isolated mild ALT <3x ULN with no symptoms — you withdraw only if ALT >3x ULN WITH symptoms/jaundice or >5x ULN asymptomatic.

8. ANA-positive quartet (drug AIH)

WHAT YOU SEE

AUTOIMMUNE HEPATITIS-LIKE quartet, ANA(+): nitrofurantoin, minocycline, hydralazine, alpha-methyldopa

WHAT IT ENCODES

Drug-induced autoimmune hepatitis presents with positive ANA/ASMA, interface hepatitis, and plasma-cell infiltrate, classically from nitrofurantoin, minocycline, hydralazine, methyldopa (also statins and anti-TNF agents). It is clinically and histologically indistinguishable from idiopathic AIH except for the drug history.

BOARD TRAP

WRONG to commit a drug-AIH patient to lifelong immunosuppression — unlike true AIH, drug-induced AIH RESOLVES on stopping the drug ($\pm$ a short steroid course) and rarely relapses after steroid withdrawal, which is the key discriminator.

9. Hepatocellular R>5 (red light)

WHAT YOU SEE

Red light HEPATOCELLULAR R>5, ALT>>AST, e.g., INH

WHAT IT ENCODES

Hepatocellular pattern = R >5, with ALT (and AST) disproportionately elevated relative to ALP. $R = (ALT/ULN) \div (ALP/ULN)$. Classic offenders: isoniazid, acetaminophen, statins, NSAIDs, methotrexate. This is the pattern that, when combined with jaundice, triggers Hy's Law.

BOARD TRAP

WRONG to reassure a hepatocellular-pattern patient who is jaundiced — hepatocellular injury PLUS jaundice is the high-mortality Hy's Law scenario, not a benign transaminitis.

10. Mixed R 2-5

WHAT YOU SEE

Yellow light MIXED R 2–5, both up

WHAT IT ENCODES

Mixed pattern = R between 2 and 5, with BOTH ALT and ALP elevated. Typical offenders include phenytoin, sulfonamides, amoxicillin-clavulanate (can be mixed or cholestatic), and carbamazepine. Often overlaps with hypersensitivity/DRESS-type idiosyncratic reactions.

BOARD TRAP

WRONG to force every drug into a single pattern — the same agent (e.g., amox-clav, phenytoin) can produce hepatocellular, mixed, OR cholestatic injury, so calculate the R-value rather than memorizing one pattern per drug.

11. Cholestatic R<2

WHAT YOU SEE

Green light CHOLESTATIC R<2, ALP/GGT up, estrogens, anabolic steroids

WHAT IT ENCODES

Cholestatic pattern = R <2, with ALP and GGT predominantly elevated, often with pruritus and jaundice. Anabolic steroids and estrogens (OCPs) cause 'bland' cholestasis without inflammation; amoxicillin-clavulanate causes mixed/cholestatic injury and is the #1 cause of cholestatic DILI overall.

BOARD TRAP

WRONG to attribute an isolated ALP rise to liver without checking GGT — a normal GGT points to a bony (or placental) ALP source, not cholestatic liver injury; GGT confirms hepatic origin.

12. Steatosis patterns

WHAT YOU SEE

MICROVESICULAR (valproate) vs MACROVESICULAR (amiodarone, methotrexate)

WHAT IT ENCODES

Microvesicular steatosis (small fat droplets, mitochondrial toxicity, often severe with lactic acidosis) = valproate, tetracyclines, NRTIs, aspirin (Reye syndrome). Macrovesicular steatosis/steatohepatitis = amiodarone, methotrexate, tamoxifen; amiodarone adds phospholipidosis with Mallory bodies mimicking alcoholic hepatitis.

BOARD TRAP

WRONG to read steatohepatitis with Mallory bodies in a non-drinker as alcoholic liver disease — amiodarone produces identical histology (plus lamellar inclusions), and the drug history is the discriminator.

13. Causality assessment detective

WHAT YOU SEE

Room 6 detective: temporal relationship, dechallenge, risk factors, exclude others, rechallenge

WHAT IT ENCODES

Causality (RUCAM) rests on: (1) temporal relation to drug start, (2) improvement on dechallenge (stopping), (3) exclusion of other causes, (4) known hepatotoxicity of the drug, and (5) recurrence on rechallenge. Dechallenge improvement is the safest and strongest practical clue.

BOARD TRAP

WRONG answer: performing a deliberate rechallenge to 'prove' the diagnosis — although positive rechallenge is the strongest causality evidence, it is contraindicated/potentially fatal and should not be intentionally done.

14. Hy's Law 10:1 sheriff

WHAT YOU SEE

HY'S LAW: severe hepatotoxicity, HY'S RATIO 10:1; ~10 with hepatocellular jaundice, 1 develops ALF

WHAT IT ENCODES

Hy's Law: drug-induced HEPATOCELLULAR injury (ALT >3x ULN) PLUS jaundice (total bilirubin >2x ULN) in the absence of biliary obstruction predicts >10% mortality or progression to acute liver failure. The classic teaching: of ~10 patients with hepatocellular jaundice, ~1 develops fatal ALF.

BOARD TRAP

WRONG to treat jaundice in hepatocellular DILI as a routine sign — it is the single best predictor of high mortality (Hy's Law), the opposite of a benign finding; an isolated rise in ALP/bilirubin without hepatocellular injury does NOT meet Hy's Law.

15. Treatment: STOP the drug first

WHAT YOU SEE

Room 7: STOP THE DRUG — first and most important step

WHAT IT ENCODES

Stopping the offending drug is the single most important and FIRST treatment step in all forms of DILI; most cases improve after dechallenge. Continued exposure after symptoms/jaundice appear drives many of the fatal cases.

BOARD TRAP

WRONG answer: continuing the drug 'to finish the course' or because LFTs 'might be unrelated' — in symptomatic or jaundiced DILI, prompt discontinuation is mandatory and lifesaving.

16. NAC specific antidote

WHAT YOU SEE

NAC firefighter — ONLY for acetaminophen overdose

WHAT IT ENCODES

N-acetylcysteine (NAC) replenishes glutathione and is the specific, proven antidote for acetaminophen toxicity; give based on the Rumack-Matthew nomogram (4-hour level) and don't delay if timing is uncertain. NAC also improves transplant-free survival in non-acetaminophen acute liver failure.

BOARD TRAP

WRONG to expect a specific antidote for idiosyncratic DILI — there is none; management is stopping the drug plus supportive care, with corticosteroids reserved for hypersensitivity/DRESS, drug-induced AIH, and checkpoint-inhibitor hepatitis.
